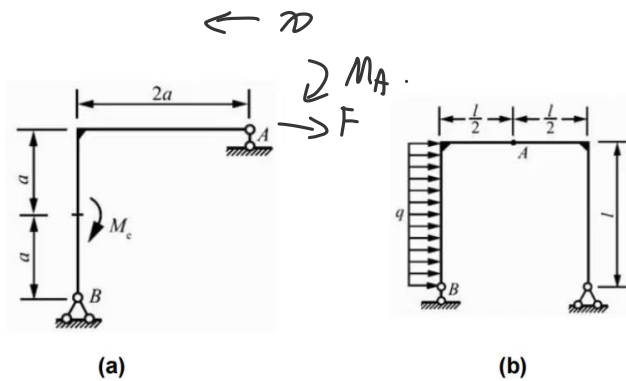


作业1/2

弯曲刚度均为EI的各刚架及其承载情况分别如图(a)~(b)所示。材料为线弹性，不计轴力和剪力的影响，试用卡氏第二定理求各刚架截面A、截面B的所有非零的位移和转角。



1) 易得 $\Delta_{Ay} = 0$.

下求 Δ_{Ax} 施加附加力 F

对AC: $M(x) = x(F + \frac{M_c}{2a})$

对CD: $M(y_1) = M_c + 2Fa - Fy_1$

对DB: $M(y_2) = Fy_2$

$$V_E = \frac{1}{2EI} \left[\int_0^{2a} (F + \frac{M_c}{2a})^2 x^2 dx + \int_0^a (M_c + 2Fa - Fy_1)^2 dy_1 + \int_0^a (Fy_2)^2 dy_2 \right]$$

$$\Delta_{Ax} = \frac{\partial V_E}{\partial F} \Big|_{F=0} = \frac{17 M_c a^2}{6EI} \quad \text{方向水平向右}$$

下求 θ_A

对AC: $M(x) = \frac{M_c + M_A}{2a} x - M_A$

对CD: $M(y_1) = M_c$

对DB: $M(y_2) = 0$

$$V_E = \frac{1}{2EI} \left(\int_0^{2a} \left[\frac{1}{2a} (M_c + M_A) x - M_A \right]^2 dx + \int_0^a M_c^2 dy_1 \right)$$

$$\theta_A = \frac{\partial V_E}{\partial M_A} \Big|_{M_A=0} = -\frac{M_c a}{3EI} \quad \text{方向顺时针}$$

(2)

对CE: $M(y_1) = -qly_1$

对AE: $M(x_1) = ql^2 - \frac{1}{2}(ql + F)x_1$

对AD: $M(x_2) = \frac{ql^2}{2} + \frac{1}{2}(ql - F)x_2$

对DB: $M(y_2) = \frac{1}{2}qy_2^2$

$$V_E = \frac{1}{2EI} \left(\int_0^l (qly_1)^2 dy_1 + \int_0^{l/2} \left[ql^2 - \frac{1}{2}(ql + x_1) \right]^2 dx_1 \right.$$

$$\left. + \int_0^{l/2} \left(\frac{1}{2}qy_2^2 \right)^2 dy_2 + \int_0^{l/2} \left[\frac{1}{2}ql^2 + \frac{1}{2}(ql - F)x_2 \right]^2 dx_2 \right)$$

$$\Delta_{Ay} = \frac{\partial V_E}{\partial F} \Big|_{F=0} = -\frac{39ql^4}{32EI} \quad \text{方向竖直向上}$$

求 Δ_{Ax} .

$$\begin{cases} CB: & M(y_1) = -(qL + F)y_1 \\ EA: & M(x_1) = (qL + F)L - (\frac{qL}{2} + F)x_1 \\ AD: & M(x_2) = \frac{qL^2}{2} + (\frac{1}{2}qL + F)x_2 \\ DB: & M(y_2) = \frac{1}{2}qy_2^2 \end{cases}$$

$$\Delta_{Ax} = \frac{\partial U_E}{\partial F} \Big|_{F=0} = \frac{3qL^4}{4EI} \quad \text{向右水平向右.}$$

求 Q_A .

$$\begin{cases} CB: & M(y_1) = -qly_1 \\ EA: & M(x_1) = qL^2 - (\frac{1}{2}qL + \frac{M_A}{L})x_1 \\ AD: & M(x_2) = \frac{qL^2}{2} + (\frac{1}{2}qL + \frac{M_A}{L})x_2 \\ DB: & M(y_2) = \frac{1}{2}qy_2^2 \end{cases}$$

$$Q_A = \frac{\partial U_E}{\partial M_A} \Big|_{M_A=0} = -\frac{qL^3}{48EI} \quad \text{逆时针.}$$

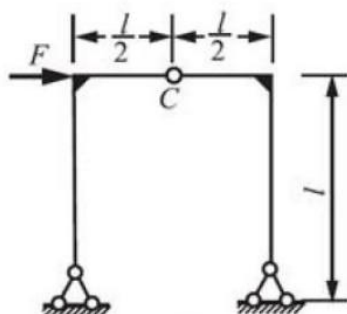
求 Q_B .

$$\begin{cases} CB: & M(y_1) = -qly_1 \\ ED: & M(x_1) = qL^2 - (\frac{1}{2}qL + \frac{M_B}{L})x_1 \\ DB: & M(y_2) = M_B - \frac{1}{2}qy_2^2 \end{cases}$$

$$Q_B = -\frac{qL^3}{24EI} \quad \text{逆时针}$$

作业2/2

弯曲刚度均为EI的各刚架及其承载情况如图所示。材料为线弹性，不计轴力和剪力的影响，试用卡氏第二定理求图示刚架上C点处两侧截面的相对角位移。



$$\text{左柱} \int_0^l \frac{1}{EI} \left(-\frac{Fy}{2} \right) \left(-\frac{y}{l} \right) dy = \frac{Fl^2}{6EI}$$

$$\text{右同理} -\frac{Fl^2}{6EI} \quad V_{\text{柱}} = \frac{Fl^2}{6EI} - \frac{Fl^2}{6EI} = 0$$

$$\text{左横梁} \int_0^{l/2} \frac{1}{EI} \left(Fx - \frac{Fl}{2} \right) (-1) dx = -\frac{Fl^2}{8EI}$$

$$\text{右为} -\frac{Fl^2}{8EI} \quad V_{\text{梁}} = -\frac{Fl^2}{8EI} - \frac{Fl^2}{8EI} = 0$$

$$AO_c = V_{\text{梁}} + V_{\text{柱}} = 0$$

→ Mc.